Fuzzy Rules & Membership Functions

Digital Detox Recommendation System

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About this document. This appendix provides the complete technical specification of the fuzzy inference system implemented in *fuzzy-detox*: all input and output membership functions with their parameters and literature justifications, and the full IF-THEN rule base for every output variable. The membership function boundaries were informed by empirical literature and operationalised as gradual fuzzy thresholds rather than fixed clinical cut-offs. The system uses a hierarchical **Mamdani** inference engine with **centroid defuzzification** across four cascaded fuzzy inference systems: FocusQuality, SleepQuality, DigitalOverload (Subsystem 1), and HabitBalance (Subsystem 2). Unlike the original three-term design, all outputs now use **five linguistic terms** (*very_ low / low / medium / medium_ high / high*), providing finer discrimination across the $[0, 10]$ output universe.

1. Input Membership Functions

All inputs are fuzzified into three linguistic terms (*Low / Medium / High*) using trapezoidal (`trapmf`) and triangular (`trimf`) membership functions.

1.1 ScreenGlances [0–150 unlocks/day]

Term	Function	Parameters
low	`trapmf`	[0, 0, 20, 40]
medium	`trimf`	[20, 50, 90]
high	`trapmf`	[65, 90, 150, 150]

The membership functions were defined to reflect the range of daily smartphone checking behaviour. The value for *medium* usage is centred around 50 glances per day, because this is close to the reported average of approximately 58 daily smartphone checks [FD]. Therefore, users around this range are interpreted as showing a typical or moderate checking pattern. The *high* membership function already begins to increase from 65 glances onward and reaches full membership around 90 glances. This choice is based on Ward et al. (2017), who reported that users glanced at their phone approximately 85 times per day [W1]. Thus, values around 85 are treated as clearly elevated, but not as an abrupt threshold.

1.2 IdleChecking [0–80 checks/day]

Term	Function	Parameters
low	`trapmf`	[0, 0, 10, 20]
medium	`trimf`	[10, 30, 55]
high	`trapmf`	[40, 58, 80, 85]

Idle checking refers to short, habitual smartphone interactions in which users briefly inspect the device without a clearly defined task goal. The lower bound represents the theoretical minimum of no idle checking, while the upper bound is chosen as a practical high-use anchor based on empirical findings on daily smartphone checking, based on Ward et al. (2017) value of 85 checks per day.

1.3 LateNightUse [0–120 minutes before bedtime]

Term	Function	Parameters
low	`trapmf`	[0, 0, 15, 30]
medium	`trimf`	[15, 30, 60]
high	`trapmf`	[31, 60, 120, 120]

LateNightUse refers to the duration of smartphone use shortly before sleep, especially smartphone use in bed. The membership functions are based on Bozkurt, Demirdöğen & Akıncı (2024) [B1], who measured adolescents' average smartphone use in bed using the categories 0–15 minutes, 16–30 minutes, 31–60 minutes, and more than 60 minutes. Therefore, the fuzzy sets are designed around these time intervals.

1.4 SocialMediaUsage [0–100% of screen time]

Term	Function	Parameters
low	`trapmf`	[0, 0, 10, 25]
medium	`trimf`	[15, 30, 55]
high	`trapmf`	[40, 70, 100, 100]

This input captures the share of total screen time devoted to social-media platforms. The membership functions are informed by recent screen-time statistics showing that social media is a highly common form of digital media use: Exploding Topics (2026) reports that 88.1% of global internet users spend screen time on social media every week, with an average of 7 hours and 6 minutes per week [ST1]. Since this statistic describes how widespread social-media use is, rather than proving that 88.1% of all screen time is spent on social media, the thresholds are used as operational design choices. The *low* function represents cases where social media forms only a smaller part of total screen time, the *medium* function captures regular but not dominant social-media use, and the *high* function begins once social media becomes the main screen-time activity.

2. Output Membership Functions

All outputs use the universe $[0, 10]$ and centroid defuzzification. Unlike the original three-term design, all four output variables now use **five terms** (*very_low* / *low* / *medium* / *medium_high* / *high*).

2.1 FocusQuality

Term	Function	Parameters
very_low	`trapmf`	[0, 0, 1.5, 3.0]
low	`trimf`	[2.0, 3.5, 5.0]
medium	`trimf`	[4.0, 5.5, 7.0]
medium_high	`trimf`	[5.5, 7.5, 9.0]
high	`trapmf`	[8.0, 9.0, 10, 10]

Ward et al. (2017) [W1] demonstrated that truly high attentional capacity requires near-total absence of digital fragmentation; this motivates the demanding *high* threshold starting at 8.0. The remaining membership functions were chosen as gradual steps across the 0–10 output scale. The overlaps between the functions reflect that focus quality does not change abruptly from one category to another, but develops gradually across the scale.

2.2 SleepQuality

Term	Function	Parameters
very_low	`trapmf`	[0, 0, 2.5, 4.5]
low	`trimf`	[3.5, 5.0, 6.5]
medium	`trimf`	[5.5, 7.0, 8.0]
medium_high	`trimf`	[7.0, 8.0, 9.0]
high	`trapmf`	[8.5, 9.5, 10, 10]

Sleep quality degrades faster with disruption than focus does, justifying the wider *very_low* zone extending to 4.5. This strict threshold reflects the assumption that very high sleep quality requires only minimal late-night phone-related stimulation. Exelmans & Van den Bulck (2016) [Ex1] show that bedtime mobile phone use is negatively associated with several sleep outcomes like sleep latency and efficiency. Therefore, the model treats truly high sleep quality as a demanding state that is less likely when late-night phone use is present.

2.3 DigitalOverload

Term	Function	Parameters
very_low	`trapmf`	[0, 0, 1.0, 2.5]
low	`trimf`	[1.5, 3.0, 4.5]
medium	`trimf`	[3.5, 5.0, 6.5]
medium_high	`trimf`	[5.5, 7.0, 8.0]
high	`trapmf`	[7.0, 8.5, 10, 10]

Sweller's (1994) [Sw1] Cognitive Load Theory argues that working memory has a limited capacity and that cognitive demands can accumulate until this capacity is exceeded. This supports the placement of the *high* overload zone starting relatively early at 7.0, because a user can already experience strong overload before reaching the extreme end of the 0–10 scale. Kushlev & Dunn (2015) [K1] further show that frequent checking behaviour is associated with higher daily stress, as participants reported lower stress when they limited email checking compared with unrestricted checking. This supports the *medium* core at 5.0.

2.4 HabitBalance (Subsystem 2 output)

HabitBalance is the final output of the hierarchical system. It is inferred by a dedicated fourth Mamdani FIS whose inputs are the defuzzified crisp values of FocusQuality, SleepQuality, and DigitalOverload. Re-injecting intermediate crisp outputs into a new FIS.

Term	Function	Parameters
very_low	`trapmf`	[0, 0, 1.5, 3.0]
low	`trimf`	[2.0, 3.5, 5.0]
medium	`trimf`	[4.0, 5.5, 7.0]
medium_high	`trimf`	[5.5, 7.0, 8.5]
high	`trapmf`	[7.0, 8.5, 10, 10]

The intermediate-input membership functions for FocusQuality_in and SleepQuality_in share the same shape: *low* = `trapmf`[0, 0, 2.5, 4.5]; *medium* = `trimf`[3.0, 5.0, 7.5]; *high* = `trapmf`[6.0, 8.0, 10, 10]. DigitalOverload_in uses the same shape but is semantically inverted in the rules (a *high* overload input drives HabitBalance downward).

3. IF-THEN Rules — FocusQuality

Rule 1.

> **IF** ScreenGlances *is High*
> **THEN** FocusQuality *is Low*

Ward et al. (2017) `[W1]` showed that even the mere presence of a smartphone reduces available cognitive capacity. Frequent screen glances directly fragment attention. `[W1]`

Rule 2.

> **IF** LateNightUse *is High*
> **THEN** FocusQuality *is Low*

Poor sleep carries over into next-day cognition through multiple mechanisms. Rasch & Born (2013) `[R1]` explain that sleep, particularly slow-wave sleep, is central to memory consolidation and cognitive restoration.

Rule 3.

> **IF** SocialMediaUsage *is High*
> **THEN** FocusQuality *is Low*

Social media platforms are designed to maximise engagement through variable reward loops, making sustained attention harder to maintain, according to Newport's Digital Minimalism. `[N2]` Kushlev & Dunn (2015) `[K1]` do not examine social media directly, but their findings show that frequent checking of digital communication, in their case email, can function as a source of task switching and is associated with higher stress.

Rule 4. [improved]

> **IF** IdleChecking *is High*
> **THEN** FocusQuality *is Low*

Siebers, Beyens & Valkenburg (2024) `[S1]` show that habitual checking creates attentional switching costs that accumulate throughout the day, which is associated with higher distraction. Therefore a standalone *high* IdleChecking input is sufficient to reduce FocusQuality to the *low* zone, but not to the *very_low* zone — that requires a co-occurring disruptor (see Rule 5). This rule was separated from the combined rule to avoid over-penalising a single elevated input. `[S1]`

Rule 5. [improved]

> **IF** IdleChecking *is High* **AND** ScreenGlances *is High*
> **THEN** FocusQuality *is Very Low*

Mark et al. (2008) `[M1]` found that it takes approximately 23 minutes to fully recover focus after an interruption. When both IdleChecking and ScreenGlances are *high*, two reinforcing fragmentation sources operate simultaneously, making meaningful recovery nearly impossible across a normal day.

Rule 6.

> **IF** ScreenGlances *is Low* **AND** SocialMediaUsage *is Low*
> **THEN** FocusQuality *is High*

Low checking frequency combined with low social media engagement defines what Newport (2016) [N1] calls a deep-work-compatible digital pattern: the phone is used intentionally and infrequently, leaving attentional resources intact for sustained cognitive effort. [N1]

Rule 7.

> **IF** ScreenGlances *is Low* **AND** IdleChecking *is Low* **AND** LateNightUse *is Low*
> **THEN** FocusQuality *is High*

All three main disruption channels are inactive — the healthiest digital pattern for focus. [N1] [R1]

Rule 8.

> **IF** ScreenGlances *is Medium* **AND** LateNightUse *is Low*
> **THEN** FocusQuality *is Medium-High*

Moderate checking frequency introduces some fragmentation, Rasch & Born (2013) [R1] describe sleep as a brain state that supports memory consolidation and note that sleep disruption is associated with cognitive and emotional problems. In the present fuzzy system, moderate IdleChecking is therefore not modelled as sufficient on its own to reduce FocusQuality to the lowest zone when SleepQuality remains intact. [K1] [R1]

Rule 9.

> **IF** ScreenGlances *is Medium* **AND** SocialMediaUsage *is Medium*
> **THEN** FocusQuality *is Medium*

Both variables are moderate — neither extreme causes collapse, but the combination keeps focus in the middle range. [S1]

Rule 10.

> **IF** ScreenGlances *is Medium* **AND** LateNightUse *is Medium*
> **THEN** FocusQuality *is Medium*

Moderate daytime checking plus moderate evening use creates a mild cumulative burden. [R1]

Rule 11.

> **IF** IdleChecking *is Medium* **AND** SocialMediaUsage *is Medium*
> **THEN** FocusQuality *is Medium*

Both sources of passive engagement — habitual checking and social media consumption — are present but not extreme. [S1]

Rule 12.

> **IF** LateNightUse *is Medium* **AND** SocialMediaUsage *is Medium*
> **THEN** FocusQuality *is Medium*

Moderate evening use with moderate social media suggests some pre-sleep stimulation, mildly affecting next-day focus. [Ex1] [R1]

Rule 13.

> **IF** ScreenGlances *is Low* **AND** SocialMediaUsage *is Medium*
> **THEN** FocusQuality *is Medium-High*

Low interruption frequency is the stronger focus protector. Newport (2016) [N1] identifies controlled checking as the primary lever for protecting attentional resources; even with medium social media engagement, low glance frequency preserves most of the focus benefit, yielding a *medium_high* outcome. [N1]

Rule 14.

> **IF** ScreenGlances *is Medium* **AND** SocialMediaUsage *is Low*
> **THEN** FocusQuality *is Medium-High*

Symmetric to Rule 13: the absence of a passive social media engagement loop compensates for some checking frequency.

Rule 15.

> **IF** IdleChecking *is Low* **AND** LateNightUse *is Medium*
> **THEN** FocusQuality *is Medium*

Moderate late-night phone use is modelled as a mild risk factor for FocusQuality because bedtime mobile phone use has been associated with poorer sleep outcomes. (Exelmans & Van den Bulck, 2016) [Ex1]. However, because IdleChecking is low, the model assumes that the user does not experience strong daytime fragmentation from habitual phone checking. This is consistent with Siebers, Beyens & Valkenburg (2024) [S1], who show that fragmented smartphone use is associated with higher distraction. Therefore, the combined profile is not treated as severely impaired, but is placed in the *Medium* FocusQuality range.

4. IF-THEN Rules — SleepQuality

Rule S1.

> **IF** LateNightUse *is Low* **AND** ScreenGlances *is Low* **AND** IdleChecking *is Low*
> **THEN** SleepQuality *is High*

The healthy baseline rule. All three disruption channels are inactive.

Rule S2.

> **IF** LateNightUse *is Low* **AND** SocialMediaUsage *is Low*
> **THEN** SleepQuality *is High*

Other healthy baseline rule.With two disruptions being inactive.

Rule S3.

> **IF** LateNightUse *is Low* **AND** ScreenGlances *is Medium*
> **THEN** SleepQuality *is Medium-High*

When late-night use is absent, moderate daytime screen glances do not carry significant sleep consequences. Exelmans & Van den Bulck (2016) [Ex1] focus specifically on bedtime mobile phone use and show that phone use in bed or after lights out is associated with poorer sleep outcomes. Therefore, when LateNightUse is absent, moderate daytime ScreenGlances are not treated as a major sleep-disrupting factor in the present fuzzy system. [Ex1]

Rule S4.

> **IF** LateNightUse *is Low* **AND** SocialMediaUsage *is Medium*
> **THEN** SleepQuality *is Medium-High*

Abdullah et al. (2025) emphasise that bedtime and nighttime social media use are especially linked with poorer sleep quality. With LateNightUse still *low*, medium SocialMediaUsage is therefore treated as a mild sleep risk rather than a strong impairment, supporting a *medium_high* SleepQuality outcome. [A2]

Rule S5.

> **IF** LateNightUse *is Medium* **AND** ScreenGlances *is Medium*
> **THEN** SleepQuality *is Medium*

Moderation rule.

Rule S6.

> **IF** LateNightUse *is Medium* **AND** SocialMediaUsage *is Medium*
> **THEN** SleepQuality *is Medium*

Moderation rule.

Rule S7.

> **IF** LateNightUse *is Medium* **AND** ScreenGlances *is Low*
> **THEN** SleepQuality *is Medium*

Exelmans and Van den Bulck (2016) show that mobile phone activity after lights out predicts poorer PSQI-related outcomes, including longer sleep latency, worse sleep efficiency, sleep disturbance, and daytime dysfunction. Since ScreenGlances are *low*, the model keeps the effect at *medium* instead of treating it as a compounded sleep-risk pattern. [Ex1]

Rule S8.

> **IF** LateNightUse *is High* **AND** SocialMediaUsage *is Medium*
> **THEN** SleepQuality *is Low*

Hjetland et al. (2025) found that each additional hour of screen time in bed was associated with 59% higher odds of insomnia symptoms and 24 minutes less sleep. Therefore, high LateNightUse combined with medium SocialMediaUsage is treated as a clear sleep-disrupting profile, placing SleepQuality in the *low* range. [HJ]

Rule S9.

IF LateNightUse *is Medium* **AND** SocialMediaUsage *is High*
THEN SleepQuality *is Low*

Abdullah et al. (2025) report that bedtime/nighttime social media use is associated with poor sleep quality. With SocialMediaUsage *high* but LateNightUse only *medium*, the rule assigns *low* SleepQuality without yet moving to *very_low*. [A2]

Rule S10.

IF ScreenGlances *is Low* **AND** LateNightUse *is High*
THEN SleepQuality *is Low*

Even when daytime ScreenGlances are controlled, persistent high late-night use significantly deteriorates sleep quality. Exelmans & Van den Bulck (2016) [Ex1] established that sleep disruption is primarily a function of bedtime proximity. [Ex1]

Rule S11.

IF LateNightUse *is High* **AND** SocialMediaUsage *is High*
THEN SleepQuality *is Very Low*

When both LateNightUse and SocialMediaUsage are *high*, the model treats the two risks as compounding and therefore assigns *very_low* SleepQuality. [H1] [A1] [R1]

Rule S12.

IF LateNightUse *is High* **AND** ScreenGlances *is High*
THEN SleepQuality *is Very Low*

Arshad et al. (2021) [Ar1] used the Pittsburgh Sleep Quality Index to demonstrate that excessive smartphone screen time combined with high late-night use leads to significantly poorer sleep across all PSQI components, yielding *very_low*. [Ar1]

Rule S13.

IF LateNightUse *is High* **AND** IdleChecking *is High*
THEN SleepQuality *is Very Low*

Siebers, Beyens and Valkenburg (2024) explain that fragmented smartphone use involves repeated attentional switching and may be driven by checking habits, while bedtime phone use is tied to poorer sleep outcomes in Exelmans and Van den Bulck (2016). In the fuzzy model, high IdleChecking therefore marks a fragmented checking pattern that can reinforce high LateNightUse, making *very_low* SleepQuality plausible. [S1] [H1]

5. IF-THEN Rules — DigitalOverload

Rule O1.

> **IF** ScreenGlances *is High* **AND** SocialMediaUsage *is High*
> **THEN** DigitalOverload *is High*

High checking frequency combined with social media as the dominant activity means the user is both frequently interrupted and spending most screen time on stimulating passive content. Mehta (2023) `[Me1]` formalised this. `[Me1]`

Rule O2.

> **IF** IdleChecking *is High* **AND** SocialMediaUsage *is High*
> **THEN** DigitalOverload *is High*

Compulsive habitual checking feeding into heavy social media use creates a reinforcing loop: the habit triggers the app, the app rewards the habit and internally triggered checks may lead into sticky app sessions, especially where feeds and autoplay encourage continued use. Siebers et al. (2024) `[S1]` describe fragmented and sticky smartphone use as patterns associated with distraction. `[S1]`

Rule O3.

> **IF** LateNightUse *is High* **AND** SocialMediaUsage *is High*
> **THEN** DigitalOverload *is High*

Evening hours dominated by social media combined with high late-night use indicates that the phone is consuming recovery time that the body would otherwise use for cognitive and physiological restoration, which means lower well-being. `[Ex1]`

Rule O4.

> **IF** ScreenGlances *is High* **AND** IdleChecking *is High* **AND** LateNightUse *is High*
> **THEN** DigitalOverload *is High*

All three principal behavioural overload channels are elevated simultaneously. Looking at Ward et al. (2017) `[W1]` and Mark et al. (2008) `[M1]` together, this is the most extreme overload combination the system can detect: every waking hour is fragmented by glances, habitual checks drain recovery windows, and late-night use eliminates the overnight restoration window. `[W1]` `[M1]`

Rule O5.

> **IF** ScreenGlances *is High* **AND** IdleChecking *is High*
> **THEN** DigitalOverload *is Medium-High*

High screen glances and high idle checking can be modelled as medium-high overload because both represent fragmented attention. This is supported by evidence that interruptions increase stress and effort. `[M1]`

Rule O6.

> **IF** IdleChecking *is High* **AND** LateNightUse *is High*
> **THEN** DigitalOverload *is Medium-High*

High idle checking plus high late-night use can be modelled as medium-high overload because it combines habitual interruption with behaviour that may reduce sleep opportunity. [S1] [Ex1]

Rule O7.

> **IF** ScreenGlances *is High* **AND** LateNightUse *is Medium*
> **THEN** DigitalOverload *is Medium-High*

High daytime fragmentation combined with moderate evening use creates a sustained overload state across most waking hours. [W1] [H1]

Rule O8.

> **IF** SocialMediaUsage *is High* **AND** ScreenGlances *is Low* **AND** LateNightUse *is Low*
> **THEN** DigitalOverload *is Medium*

Improvement note. Originally, any *high* SocialMediaUsage input alone triggered *high* DigitalOverload. In the revised rule base, a single elevated input is treated as a partial risk rather than a maximum-risk profile. Because ScreenGlances and LateNightUse are both *low*, the model assumes that social media use is dominant within screen time but not accompanied by frequent interruptions or late-night recovery loss. Therefore, the consequent is reduced to *medium* DigitalOverload.

Rule O9.

> **IF** LateNightUse *is High* **AND** SocialMediaUsage *is Low*
> **THEN** DigitalOverload *is Medium*

Late-night use without heavy social media engagement is less likely to involve stimulating variable-reward content. However, High late-night use can still create digital strain because electronic media use close to sleep is associated with poorer sleep quality and more sleep-related problems. (Han et al., 2024 [H1]). Therefore, digital overload is classified as *Medium* rather than *High*.

Rule O10.

> **IF** ScreenGlances *is High* **AND** SocialMediaUsage *is Low*
> **THEN** DigitalOverload *is Medium*

Frequent screen glances can still increase digital strain because repeated checking interrupts attention and is associated with higher stress. Kushlev & Dunn (2015) [K1] show that limiting email checking to three times per day reduced daily stress compared with unrestricted checking, which supports a *Medium* overload level when checking is frequent but not dominated by social media.

Rule O11.

> **IF** IdleChecking *is High* **AND** SocialMediaUsage *is Low*
> **THEN** DigitalOverload *is Medium*

High idle checking suggests a stress pattern caused by repeated monitoring rather than by long social-media consumption. Kushlev & Dunn (2015) [K1] provide experimental evidence that checking email less frequently lowers daily stress, so frequent checking alone justifies digital overload, while low social media use keeps the outcome below *High*.

Rule O12.

> **IF** ScreenGlances *is Medium* **AND** SocialMediaUsage *is Medium*
> **THEN** DigitalOverload *is Medium*

Two moderate inputs represent the most common everyday digital pattern observed in population samples. Mehta (2023) [Me1] characterises this as the modal overload profile — neither particularly healthy nor severely harmful, representing the Goldilocks zone also described by Przybylski & Weinstein (2017) [P1]. [Me1] [P1]

Rule O13.

> **IF** ScreenGlances *is Low* **AND** SocialMediaUsage *is High*
> **THEN** DigitalOverload *is Medium*

Heavy social media despite low checking frequency suggests longer focused scrolling sessions rather than many short ones. While less disruptive to attentional flow (Ward et al., 2017 [W1]), the content type still carries overload risk through the emotional activation that Ahmed et al. (2024) [A1] link to heavy social media engagement. [W1] [A1]

Rule O14.

> **IF** ScreenGlances *is Medium* **AND** SocialMediaUsage *is Low* **AND** LateNightUse *is Low*
> **THEN** DigitalOverload *is Low*

Moderate checking with low social media and no evening engagement represents a largely intentional-use pattern. The combination of moderate frequency and low passive content suggests a compatible with a low-overload lifestyle. [N1]

Rule O15.

> **IF** IdleChecking *is Medium* **AND** ScreenGlances *is Low*
> **THEN** DigitalOverload *is Low*

Medium idle checking with low screen glances can be modelled as low overload because the habitual-checking signal is present but not paired with frequent visible interruptions. Siebers supports the relevance of fragmented use for distraction, but the low glance input weakens the rule. [S1]

Rule O16.

> **IF** ScreenGlances *is Low* **AND** SocialMediaUsage *is Low*
> **THEN** DigitalOverload *is Very Low*

Both the fragmentation source and the passive content source are absent. Newport (2016) [N2] describes this as intentional, low-stimulation use — the pattern most consistent with digital minimalism and the lowest achievable overload state under typical modern phone ownership. [N2]

Rule O17.

> **IF** ScreenGlances *is Low* **AND** IdleChecking *is Low* **AND** LateNightUse *is Low*
> **THEN** DigitalOverload *is Very Low*

All three principal overload channels are inactive — the healthy digital baseline.

6. IF-THEN Rules — HabitBalance (Subsystem 2)

6.1 Cluster 1 — Focus High and Sleep High (dominant positive signals)

Newport (2016) [N1] argues in *Deep Work* that minimizing digital distractions is essential for protecting our finite attentional capacity, a theme expanded to broader lifestyle choices in his later work (2019). Ward et al. (2017) [W1] demonstrate that protecting this cognitive capacity from environmental digital drains is a critical component of baseline cognitive performance. These rules therefore represent the best-case corner of the input space.

Rule H1.

> **IF** FocusQuality *is High* **AND** SleepQuality *is High* **AND** DigitalOverload *is Low*
> **THEN** HabitBalance *is High*

All positive signals are maximised and no overload is present. This configuration aligns with Newport's (2016) framework of an optimized cognitive environment: high daytime focus unencumbered by digital fragmentation, allowing for deep execution, while low overload prevents the passive "brain drain" identified by Ward et al. (2017) [W1]. [N1] [W1]

Rule H2.

> **IF** FocusQuality *is High* **AND** SleepQuality *is High* **AND** DigitalOverload *is Medium*
> **THEN** HabitBalance *is Medium-High*

High focus and sleep provide strong positive signals; the moderate overload introduces a partial but not critical drain. Newport (2016) [N1] notes that some digital engagement is unavoidable in modern life, its disruptive impact is minimized when core attentional habits and recovery baselines remain intact. [N1]

Rule H3.

> **IF** FocusQuality *is High* **AND** SleepQuality *is High* **AND** DigitalOverload *is High*
> **THEN** HabitBalance *is Medium*

Even strong focus and sleep cannot fully compensate for a high overload state. Ward et al. (2017) [W1] demonstrate that the mere presence of highly engaging digital devices exerts an independent tax on available cognitive capacity, capping the overall HabitBalance at a *medium* level despite positive personal inputs. [W1]

6.2 Cluster 2 — Overload High as a nullifying factor

Sweller (2010) [Sw1] establishes that working memory has a capacity limit; when extraneous load is excessive, cognitive processing efficiency drops. Kushlev & Dunn (2015) [K1] show that checking email less frequently reduces daily stress, suggesting that frequent digital checking can contribute to psychological strain, even if its broader effects on well-being are more indirect.

Rule H4.

> **IF** FocusQuality *is High* **AND** SleepQuality *is Low* **AND** DigitalOverload *is High*
> **THEN** HabitBalance *is Very Low*

Although focus quality is high, low sleep quality and high digital overload create two strong negative conditions for habit balance. Sweller (1994) [Sw1] explains that limited working-memory resources can be impaired when cognitive demands are excessive, while Rasch & Born (2013) [R1] show that sleep is important for memory consolidation and cognitive restoration. Therefore, the rule treats high focus as insufficient to compensate for the combined burden of poor sleep and high overload. [Sw1] [R1]

Rule H5.

> **IF** FocusQuality *is Medium* **AND** SleepQuality *is Medium* **AND** DigitalOverload *is High*
> **THEN** HabitBalance *is Low*

When focus quality and sleep quality are only moderate, high digital overload becomes the dominant negative signal. Kushlev & Dunn (2015) [K1] show that checking email less frequently can reduce daily stress, suggesting that frequent digital checking may add psychological strain even when other aspects of functioning are not severely impaired. For this reason, the rule assigns the overall habit balance to the *low* range. [K1]

Rule H6.

> **IF** FocusQuality *is Low* **AND** SleepQuality *is Low* **AND** DigitalOverload *is High*
> **THEN** HabitBalance *is Very Low*

The worst-case overload-dominated scenario. Both positive inputs are depleted and cognitive load is maximal. [Sw1] [K1]

6.3 Cluster 3 — Sleep Low as an independent aggravating factor

Low sleep quality is treated as an independent aggravating factor because sleep supports memory consolidation and broader cognitive functioning. Rasch & Born (2013) [R1] provide the mechanistic grounding by describing the role of sleep, especially SWS, in consolidation processes. For the performance-related argument, Van Dongen et al. (2003) [VD1] show that repeated sleep restriction can produce cumulative neurobehavioural deficits. [R1] [VD1]

Rule H7.

> **IF** FocusQuality *is High* **AND** SleepQuality *is Low* **AND** DigitalOverload *is Low*
> **THEN** HabitBalance *is Medium*

High focus and low overload protect the system from falling into a clearly negative state. However, low sleep quality remains a limiting factor because sleep contributes to memory consolidation and cognitive recovery (Rasch & Born, 2013 [R1]). The rule therefore assigns a *medium* outcome: the user still has strong focus resources, but poor sleep prevents a high HabitBalance. [R1]

Rule H8.

> **IF** FocusQuality *is Medium* **AND** SleepQuality *is Low* **AND** DigitalOverload *is Medium*
> **THEN** HabitBalance *is Low*

Here, neither positive input is strong enough to offset the sleep deficit. Rasch & Born (2013) [R1] show that sleep is central for consolidation processes, and Van Dongen et al. (2003) [VD1] demonstrate that restricted sleep can accumulate into measurable neurobehavioural impairment. With only medium focus and medium overload, the system lacks a strong compensating factor, so the outcome falls into the *low* range. [R1] [VD1]

Rule H9.

> **IF** FocusQuality *is Low* **AND** SleepQuality *is Low* **AND** DigitalOverload *is Low*
> **THEN** HabitBalance *is Low*

Both primary wellbeing indicators are depleted. Even with low overload, the absence of restorative sleep (Rasch & Born, 2013 [R1]) and attentional capacity leaves the system in a *low* balance state. [R1]

6.4 Cluster 4 — Goldilocks zone (partial compensations)

Przybylski & Weinstein (2017) [P1] propose the digital Goldilocks hypothesis, according to which moderate digital engagement is not intrinsically harmful and may even be compatible with wellbeing, whereas excessive use can become problematic when it displaces other meaningful activities. Rules in this cluster capture mixed constellations in which no single input fully dominates the system, leading mainly to *medium* or *low* HabitBalance outcomes.

Rule H10.

> **IF** FocusQuality *is Medium* **AND** SleepQuality *is Medium* **AND** DigitalOverload *is Low*
> **THEN** HabitBalance *is Medium*

This rule represents a balanced but not optimal state. Medium focus and medium sleep provide sufficient positive resources, while low overload prevents additional strain. In line with the Goldilocks hypothesis, moderate digital engagement is treated as neither clearly harmful nor strongly beneficial, which justifies a *medium* rather than high HabitBalance. [P1]

Rule H11.

> **IF** FocusQuality *is Medium* **AND** SleepQuality *is Medium* **AND** DigitalOverload *is Medium*
> **THEN** HabitBalance *is Medium*

The fully moderate-all case. No dimension dominates, and the system sits at a stable but unexceptional equilibrium. [P1]

Rule H12.

> **IF** FocusQuality *is High* **AND** SleepQuality *is Low* **AND** DigitalOverload *is Medium*
> **THEN** HabitBalance *is Low*

High focus is a strong positive signal, but low sleep quality limits the system's capacity to recover and maintain stable cognitive functioning. Rasch & Born (2013) [R1] describe sleep as important for memory consolidation, while Van Dongen et al. (2003) [VD1] show that restricted sleep can produce cumulative neurobehavioural performance deficits. Because medium overload adds further pressure, the rule treats high focus as insufficient to compensate for poor sleep, resulting in a *low* HabitBalance. [R1] [VD1]

6.5 Remaining Rules (complete coverage of all 27 combinations)

The following rules complete the full enumeration of $3^3 = 27$ logical input combinations. They do not introduce new theoretical mechanisms, but ensure that the Mamdani system remains fully defined across the entire input space. The same principle established in the previous clusters is applied consistently: higher focus and sleep quality increase HabitBalance, while higher digital overload reduces it. These rules are therefore mainly included for technical completeness and to avoid uncovered or unstable areas in the fuzzy output surface.

Rule H13.

> **IF** FocusQuality *is High* **AND** SleepQuality *is Medium* **AND** DigitalOverload *is Low*
> **THEN** HabitBalance *is Medium-High*

Rule H14.

> **IF** FocusQuality *is High* **AND** SleepQuality *is Medium* **AND** DigitalOverload *is Medium*
> **THEN** HabitBalance *is Medium*

Rule H15.

> **IF** FocusQuality *is High* **AND** SleepQuality *is Medium* **AND** DigitalOverload *is High*
> **THEN** HabitBalance *is Low*

Rule H16.

> **IF** FocusQuality *is Medium* **AND** SleepQuality *is High* **AND** DigitalOverload *is Low*
> **THEN** HabitBalance *is Medium-High*

Rule H17.

> **IF** FocusQuality *is Medium* **AND** SleepQuality *is High* **AND** DigitalOverload *is Medium*
> **THEN** HabitBalance *is Medium*

Rule H18.

> **IF** FocusQuality *is Medium* **AND** SleepQuality *is High* **AND** DigitalOverload *is High*
> **THEN** HabitBalance *is Low*

Rule H19.

> **IF** FocusQuality *is Medium* **AND** SleepQuality *is Low* **AND** DigitalOverload *is Low*
> **THEN** HabitBalance *is Low*

Rule H20.

> **IF** FocusQuality *is Medium* **AND** SleepQuality *is Low* **AND** DigitalOverload *is High*
> **THEN** HabitBalance *is Very Low*

Rule H21.

> **IF** FocusQuality *is Low* **AND** SleepQuality *is High* **AND** DigitalOverload *is Low*
> **THEN** HabitBalance *is Medium*

Rule H22.

> **IF** FocusQuality *is Low* **AND** SleepQuality *is High* **AND** DigitalOverload *is Medium*
> **THEN** HabitBalance *is Low*

Rule H23.

IF FocusQuality *is Low* **AND** SleepQuality *is High* **AND** DigitalOverload *is High*
THEN HabitBalance *is Very Low*

Rule H24.

IF FocusQuality *is Low* **AND** SleepQuality *is Medium* **AND** DigitalOverload *is Low*
THEN HabitBalance *is Low*

Rule H25.

IF FocusQuality *is Low* **AND** SleepQuality *is Medium* **AND** DigitalOverload *is Medium*
THEN HabitBalance *is Low*

Rule H26.

IF FocusQuality *is Low* **AND** SleepQuality *is Medium* **AND** DigitalOverload *is High*
THEN HabitBalance *is Very Low*

Rule H27.

IF FocusQuality *is Low* **AND** SleepQuality *is Low* **AND** DigitalOverload *is Medium*
THEN HabitBalance *is Very Low*

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